

USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Title: Lane Margining Optional Additions Applied to: USB4 Specification Version 2.0

Brief description of the functional changes:
Adding and optional feature to run Lane Margining on the corresponding Port of a Re-timer. Adding optional extension for the error counters to support long duration tests.

Benefits as a result of the changes:
Will enable Lane Margining on “other” side of the Re-timer. Will give a more accurate result of the TER.

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
None

An analysis of the hardware implications:
None

An analysis of the software implications:
None

An analysis of the compliance testing implications:
None

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Actual Change

(a). Table 8-81 - READ_LANE_MARGIN_CAP Completion Data

To Text:

DW	Bit(s)	Field Name and Description
0	1:0	Gen 2/3 Receiver Lane Margining Modes This field indicates the Receiver Lane Margining modes supported by the USB4 Port for Gen 2 and Gen 3 Links. 00b: Reserved 01b: Only HW mode supported 10b: Only SW mode supported 11b: Both HW mode and SW mode supported
	2	Multiple-Lane Margining This field indicates whether the USB4 Port supports simultaneous Lane Margining on both Lanes. 0b: Each Lane is tested separately 1b: All Lanes can be tested simultaneously
	4:3	Gen 2/3 Independent High/Low Voltage Margin This field indicates how Gen 2/3 voltage margins are returned: 00b: The Router returns the minimum between the high and low voltage margins 01b: The Router returns either the high voltage margin or the low voltage margin 10b: The Router returns both the high voltage margin and the low voltage margin (relevant only for HW mode) 11b: Reserved
	5	Gen 2/3 Time Margining This field indicates if Time Margining is supported. 0b: Time Margining is not supported for Gen 2/3 1b: Time Margining is supported for Gen 2/3
	12:6	Gen 2/3 Voltage Margin Steps – Mandatory Range This field contains the number of voltage margin steps that are supported by the USB4 Port. The number of margin steps applies to both the high margin and the low margin separately. A Router shall set this value to a minimum of 25.
	18:13	Gen 2/3 Maximum Voltage Offset – Mandatory Range This field contains the maximum supported voltage offset for Gen 2 and Gen 3 Links. The maximum voltage offset applies to both the high margin and the low margin separately. A value of n in this field corresponds to a maximum voltage offset of $[74 + n \times 2]$ mV.
	19	Gen 2/3 Optional Voltage Offset Range This field indicates if the USB4 Port supports an optional voltage offset range. 0b: Optional Voltage Offset Range is not supported 1b: Optional Voltage Offset Range is supported
	26:20	Gen 2/3 Voltage Margin Steps – Optional Range This field contains the number of voltage margin steps that are supported by the USB4 Port for the Optional Voltage Offset Range. The number of margin steps applies to both the high margin and the low margin separately. A Router shall set this value to a minimum of 25. This field is reserved if the <i>Optional Voltage Offset Range</i> field is 0b.

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DW	Bit(s)	Field Name and Description
	<u>27</u>	<u>Corresponding Port Supported</u> <u>This field indicate if a USB4 Re-timer supports the activation of Lane Margining on the corresponding Port by executing the Port Operation this Port. See <i>Port Select</i> field in Table 8-82 and Table 8-85.</u> <u>0b: Executing Lane Margining on the Corresponding Port is not supported.</u> <u>1b: Executing Lane Margining on the Corresponding Port is supported.</u> <u>For a USB4 Router, this bit shall be set to 0b.</u>
	<u>28</u>	<u>Extended Error Counter Support</u> <u>This field indicate if a USB4 Router/Re-timer supports an extended error counter when using SW Lane Margining. See <i>Extended Error Counter</i> field in Table 8-87.</u> <u>0b: Extended Error Counter is not supported.</u> <u>1b: Extended Error Counter is supported.</u>
	31: <u>29</u> 7	Reserved

(b). Table 8-82 - RUN_HW_LANE_MARGINING Operation Metadata
To Text:

DW	Bit(s)	Field Name and Description
0	2:0	Lane Select This field indicates which Lane(s) to perform the Receiver Lane Margining test on: 000b: RX0 001b: RX1 010b: RX2 (only applies to a Port that supports Asymmetric Link with 3 Rx) 111b: Test runs on all Lanes Other values are Reserved. The Port Operation shall fail if this field is set to 111b and the Router supports Lane Margining on a single Lane only as present in the <i>Two-Lane Margining</i> bit.
	3	Timing Margin Test This bit selects between a voltage margin test and a timing margin test 0b: Perform a voltage margin test 1b: Perform a timing margin test The Port Operation shall fail if this field is set to 1b and the Router does not support timing margin testing.

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DW	Bit(s)	Field Name and Description
	4	Enable Margin Tests For a Gen 2/3 voltage margin test, this field enables high margin test or low margin test when the <i>Gen 2/3 Independent High/Low Voltage Margin</i> field in the READ_LANE_MARGIN_CAP Completion Metadata was set to 01b: 0b: Enables low margin test 1b: Enables high margin test For a Gen 4 voltage margin test, this field chooses the lower or upper eye when the <i>Gen 4 Independent High/Low Voltage Margin</i> field in the READ_LANE_MARGIN_CAP Completion Metadata was set to 01b: 0b: Enables margin test on lower eye 1b: Enables margin test on upper eye For a Gen 2/3 timing margin test, this field enables right margin test or left margin test when the <i>Gen 2/3 Independent Left/Right Timing Margin Test</i> field in the READ_LANE_MARGIN_CAP Completion Metadata was set to 01b: 0b: Enables left margin test 1b: Enables right margin test For a Gen 4 timing margin test, this field chooses the lower or upper eye when the <i>Gen 4 Independent Left/Right Timing Margin</i> field in the READ_LANE_MARGIN_CAP Completion Metadata was set to 01b: 0b: Enables margin test on lower eye 1b: Enables margin test on upper eye
	9:5	BER Level Contour This field sets the BER level contour: For even values n, the maximum BER level contour supported is $1e[-12 + n/2]$. For odd values n, the maximum BER level contour supported is $3 \times 1e[-12 + (n-1)/2]$. Values larger than 10h are Reserved.
	10	Enable Optional Voltage Offset Range For a Gen 2/3 voltage margin test, this field enables the mandatory voltage margin offset range or the optional extended voltage offset range when the <i>Gen 2/3 Optional Voltage Offset Range</i> field in the READ_LANE_MARGIN_CAP Completion Metadata was set to 1b: 0b: Enables the mandatory Rx voltage offset range 1b: Enables the optional Rx voltage offset range This bit is reserved for Gen 4.
	<u>11</u>	<u>Port Select</u> <u>This field indicate on which Port the Lane Margining is activated:</u> 0b: <u>Activate Lane Margining on this Port</u> 1b: <u>Activate Lane Margining on the Corresponding Port</u> <u>When using the Operation on a USB4 Router, this bit shall be set to 0b</u>
	31:12+ 11	Reserved

(c). Table 8-85 - RUN_HW_LANE_MARGINING Completion Data

To Text:

DW	Bit(s)	Field Name and Description
0	2:0	Lane Select This field contains the value of the <i>Lane Select</i> field in the last RUN_HW_LANE_MARGINING Port Operation.
	3	Timing Margin Test This field contains the value of the <i>Timing Margin Test</i> field in the last RUN_HW_LANE_MARGINING Port Operation.

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DW	Bit(s)	Field Name and Description
	4	Enable Margin Tests This field contains the value of the <i>Enable Margin Tests</i> field in the last RUN_HW_LANE_MARGINING Port Operation.
	9:5	BER Level Contour This field contains the value of the <i>BER Level Contour</i> field in the last RUN_HW_LANE_MARGINING Port Operation.
	10	Enable Optional Voltage Offset Range This field contains the value of the <i>Enable Optional Voltage Offset Range</i> field in the last RUN_HW_LANE_MARGINING Port Operation.
	<u>11</u>	<u>Port Select</u> <u>This field contains the value of the <i>Port Select</i> field in the last RUN_HW_LANE_MARGINING Port Operation.</u>
	31:1 <u>2</u> 1	Reserved.

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(d). Table 8-85 - RUN_SW_LANE_MARGINING Operation Metadata To Text:

DW	Bit(s)	Field Name and Description
0	2:0	Lane Select This field indicates which Lane(s) to perform the Receiver Lane Margining test on: 000b: RX0 001b: RX1 010b: RX2 (only applies to a Port that supports Asymmetric Link with 3 Rx) 111b: Test runs on all Lanes Other values are Reserved. The Port Operation shall fail if this field is set to 111b and the Router supports Lane Margining on a single Lane only as present in the <i>Two-Lane Margining</i> bit.
	3	Timing Margin Test This bit selects between a voltage margin test and a timing margin test: 0b: Perform a voltage margin test 1b: Perform a timing margin test The Port Operation shall fail if this field is set to 1b and the Router does not support timing margin testing.
	4	Enable Margin Tests For a voltage margin test, this field enables high margin test or low margin test when the <i>Gen 2/3 Independent High/Low Voltage Margin</i> or the <i>Gen 4 Independent High/Low Voltage Margin</i> field in the READ_LANE_MARGIN_CAP Completion Metadata was set to 01b: 0b: Enables low margin test 1b: Enables high margin test For a timing margin test, this field enables right margin test or left margin test when the <i>Gen 2/3 Independent Left/Right Timing Margin Test</i> or the <i>Gen 4 Independent Left/Right Timing Margin Test</i> field in the READ_LANE_MARGIN_CAP Completion Metadata was set to 01b: 0b: Enables left margin test 1b: Enables right margin test
	5	Enable Optional Voltage Offset Range For a Gen 2/3 voltage margin test, this field enables the mandatory voltage margin offset range or the optional extended voltage offset range when the <i>Optional Voltage Offset Range</i> field in the READ_LANE_MARGIN_CAP Completion Metadata was set to 1b: 0b: Enables the mandatory Rx voltage offset range 1b: Enables the optional Rx voltage offset range This bit is reserved for Gen 4.
	12:6	Voltage Offset / Time Offset In a voltage margin test, this field sets the number of voltage offset steps. In a timing margin test, this field sets the number of time offset steps. Values larger than 1Fh are Reserved when performing a timing margin test.
	14:13	Error Counter 00b: NOP – no change in counter setup 01b: CLEAR – set the error counter to 0 and enable counter 10b: START – start counter, continue counting from last value 11b: STOP – stop counter, do not clear value This field affects the Error Counter for the Lanes that perform the test as selected by the <i>Lane Select</i> field. This field is Reserved if the target of the Port Operation supports destructive Time Margining.

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DW	Bit(s)	Field Name and Description
	15	Select PAM3 Eye For a voltage / timing margin test on Gen 4 Link, this field selects the eye on which the measurement is done.: 0b: Measurement shall be done on lower eye 1b: Measurement shall be done on upper eye This field is Reserved if the Link is not Gen 4.
	<u>16</u>	Port Select <u>This field indicate on which Port the Lane Margining is activated:</u> 0b: Activate Lane Margining on this Port 1b: Activate Lane Margining on the Corresponding Port <u>When using the Operation on a USB4 Router, this bit shall be set to 0b</u>
	31:1 <u>76</u>	Reserved

(e). Table 8-86 - RUN_SW_LANE_MARGINING Completion Data
To Text:

DW	Bit(s)	Field Name and Description
0	2:0	Lane Select This field contains the value of the <i>Lane Select</i> field in the last RUN_SW_LANE_MARGINING Port Operation.
	3	Timing Margin Test This field contains the value of the <i>Timing Margin Test</i> field in the last RUN_SW_LANE_MARGINING Port Operation.
	4	Enable Margin Tests This field contains the value of the <i>Enable Margin Tests</i> field in the last RUN_SW_LANE_MARGINING Port Operation.
	5	Enable Optional Voltage Offset Range This field contains the value of the <i>Enable Optional Voltage Offset Range</i> field in the last RUN_SW_LANE_MARGINING Port Operation.
	12:6	Voltage Offset / Time Offset This field contains the value of the <i>Voltage Offset / Time Offset</i> field in the last RUN_SW_LANE_MARGINING Port Operation.
	<u>13</u>	Port Select <u>This field contains the value of the <i>Port Select</i> field in the last RUN_SW_LANE_MARGINING Port Operation.</u>
	31:1 <u>43</u>	Reserved

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(f). Table 8-87 - READ_SW_MARGIN_ERR Completion Metadata To Text:

DW	Bit(s)	Field Name and Description
0	3:0	Error Counter (RX0) This field contains the number of samples with values below the voltage offset measured on RX0. The counter value increments from 0 and <i>if Extended Error Counter is not supported</i> , shall stop counting at 0Fh.
	7:4	Error Counter (RX1) This field contains the number of samples with values below the voltage offset measured on RX1. The counter value increments from 0 and <i>if Extended Error Counter is not supported</i> , shall stop counting at 0Fh.
	11:8	Error Counter (RX2) This field contains the number of samples with values below the voltage offset measured on RX2. The counter value increments from 0 and <i>if Extended Error Counter is not supported</i> , shall stop counting at 0Fh.
	<u>31:12</u>	<u>Reserved</u>
<u>1</u>	<u>11:0</u>	<u>Extended Error Counter (RX0)</u> <i>If supported (see Extended Error Counter Support field in Table 8-81), this field contain the 12 MSB of the Error Counter for RX0. Together with Error Counter (RX0) it contains the number of samples with values below the voltage offset measured on RX0. The counter shall stop counting at FFFFh.</i>
	<u>31:12</u>	<u>Reserved</u>
<u>2</u>	<u>11:0</u>	<u>Extended Error Counter (RX1)</u> <i>If supported (see Extended Error Counter Support field in Table 8-81), this field contain the 12 MSB of the Error Counter for RX1. Together with Error Counter (RX1) it contains the number of samples with values below the voltage offset measured on RX1. The counter shall stop counting at FFFFh.</i>
	<u>31:12</u>	<u>Reserved</u>
<u>3</u>	<u>11:0</u>	<u>Extended Error Counter (RX2)</u> <i>If supported (see Extended Error Counter Support field in Table 8-81), this field contain the 12 MSB of the Error Counter for RX2. Together with Error Counter (RX2) it contains the number of samples with values below the voltage offset measured on RX2. The counter shall stop counting at FFFFh.</i>
	31:12	Reserved